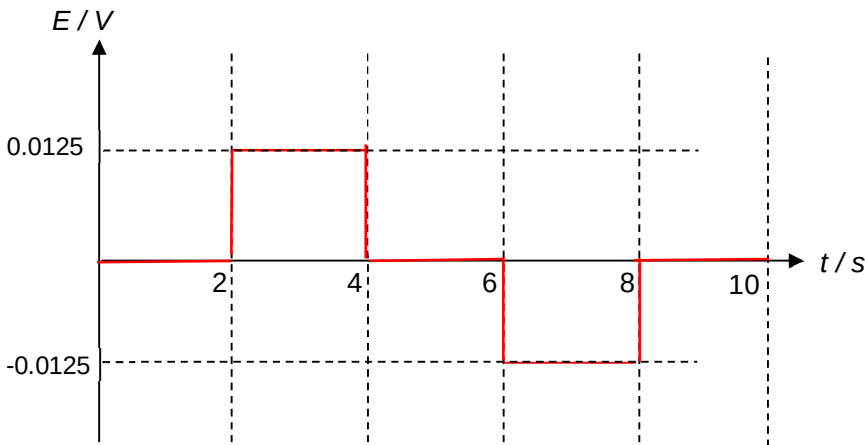


|   |     |      |                                                                                                                                                            |          |
|---|-----|------|------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 5 | (a) | (i)  | The magnetic flux through a plane surface is the <u>product of the magnetic flux density normal to the surface and the area of the surface.</u>            | B1       |
|   |     | (ii) | The magnetic flux linkage through a coil of $N$ turns is the <u>product of the number of turns of the coil and the magnetic flux</u> linking each turn.    | B1       |
|   | (b) | (i)  | $ \varepsilon  = \frac{d\phi}{dt} = \frac{d(NAB)}{dt} = \frac{d(N\ell^2 B)}{dt} = NB\ell v$ $= 5 \times 2.0 \times 0.05 \times 0.025$ $= 0.0125 \text{ V}$ | C1<br>A1 |

|  |  |       |                                                                                                                                                                                                                                                                                                                                                        |              |
|--|--|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
|  |  | (ii)  |  <p>M1 for correct shape<br/>A1 for correct label of <math>E_{\max}</math> value<br/>Allow -0.0125 V for <math>2 \text{ s} \leq t \leq 4 \text{ s}</math> and +0.0125 V for <math>6 \text{ s} \leq t \leq 8 \text{ s}</math></p>                                     |              |
|  |  | (iii) | <p>The (larger) <u>eddy currents induced</u> or <u>larger/more induced current</u> in the copper disc. [B1]</p> <p>Will result in a <u>larger force that opposes the direction of motion</u><br/>Or will result in <u>more energy dissipated as heat.</u> [B1]</p> <p>thereby requiring more energy to maintain the copper disc at constant speed.</p> | B1<br><br>B1 |

|   |     |     |  |
|---|-----|-----|--|
| 6 | (a) | (i) |  |
|---|-----|-----|--|